

Disease Detective: Investigating a Simulated Outbreak

CDC Solve the Outbreak · CDC

Senior Secondary (Years 11-12)

~30 minutes

Science · Biology

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://www.cdc.gov/mobile/applications/sto/web-app.html>

Curriculum links: disease · public health · forensic science

Learning intentions

- I can describe the process epidemiologists use to investigate a disease outbreak.
- I can evaluate evidence to identify a likely disease agent and its source.
- I can justify a public health recommendation using evidence gathered during an investigation.

Getting started

Open CDC Solve the Outbreak and start the first scenario. Read the initial briefing carefully and use the in-app tools (interviews, lab results, maps) to gather clues as you work through each stage. Keep brief notes on the evidence you collect as you go, since you'll need it for the tasks below.

Tasks

1. Summarise the initial outbreak scenario in 2-3 sentences: what symptoms are patients showing, and where is the outbreak occurring?

2. List the first three pieces of evidence or data you chose to investigate, and explain why you investigated them in that order.

3. Based on the symptoms and evidence gathered so far, list at least two possible causes (pathogens or contamination sources) you are considering, and the evidence for each.

4. As you gather more evidence in the simulation, explain how you narrowed down from your initial suspects to the most likely cause. What piece of evidence was most convincing, and why?

5. Identify the confirmed (or most likely) source of the outbreak in your scenario. How did investigators trace it back to this source?

6. Public health teams often have to act before they are 100% certain of the cause. Explain one action taken (or that you would recommend) in the scenario to protect the public while the investigation is still ongoing.

7. Evaluate the investigation process: which piece of evidence do you think was least reliable or most open to a different interpretation? Explain your reasoning.

8. Write a short summary (as if for a public health report) explaining what caused the outbreak, how it spread, and what should be done to prevent a similar outbreak in future.

Extension (optional)

Real disease outbreaks often involve incomplete or conflicting evidence, unlike the guided clues in this simulation. Choose one part of your investigation and describe what additional real-world data (e.g. genomic testing, supply chain records) might have helped confirm your conclusion faster.
